Data Science and Visualization

Unit I

FOUNDATIONS OF DATA SCIENCE

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Unit I – Foundations of Data Science

Defining Data Science - The Data Science Process - Data Types and Attributes - Python Essentials: NumPy & Pandas - Data Loading and Inspection - Data Visualization Goals - Introduction to Matplotlib & Seaborn - Basic Plotting and Customization.

Data Visualization Goals

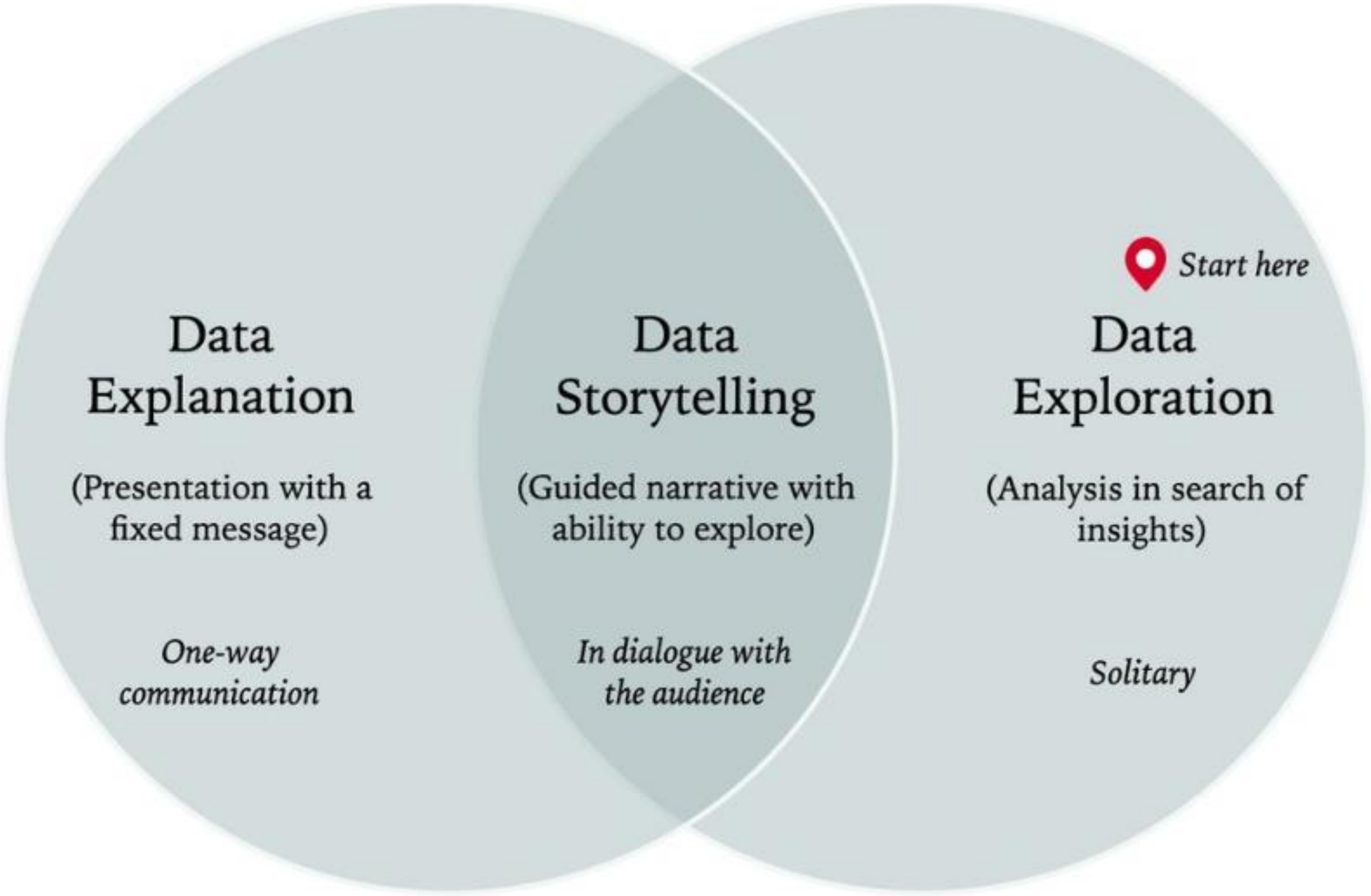
Data Visualization: Focus on the Outcome, Not the Tool

Data Visualization Goals

- Data visualization goals are the **specific objectives a designer aims to achieve** when converting raw information into a visual format.

- These goals vary depending on the audience and the complexity of the data, but they generally fall into three core categories:
  - ✓ Exploration
  - ✓ Monitoring
  - ✓ Explanation

Data Visualization Goals



1. Primary Strategic Goals

- To Explore (Discovery): **Used when patterns and insights are not yet known**. Interactive tools allow users to filter, drill down, and iterate quickly to find hidden relationships, clusters, or anomalies within complex datasets.

- To Monitor (Performance): Dashboards are designed for this goal, **providing real-time or periodic updates on key performance indicators (KPIs).** This allows stakeholders to track progress against benchmarks and take immediate action when data deviates from expectations.

- To Explain (Communication): Often called **"storytelling with data,"** this goal focuses on guiding a broad audience through a specific narrative or **"why" behind a problem.** These visualizations are hand-crafted to simplify complex concepts and drive clear conclusions.

2. Functional Objectives

Beyond the high-level strategy, effective data visualization serves several functional purposes:

- Simplification: Turning "noise" into "signal" by making large or abstract datasets accessible and digestible for non-technical users.
- Identifying Trends and Outliers: Visualizing data over time or across categories makes it instantly easier to see growth patterns, seasonality, or atypical phenomena that would be missed in a spreadsheet.
- Speeding Up Decision-Making: By leveraging human visual perception, visualizations allow decision-makers to grasp key takeaways in seconds, reducing the time spent manually scanning numbers.
- Clarifying Relationships: Visuals like scatter plots or network diagrams highlight correlations and dependencies between different variables, revealing how one factor influences another.

3. Design Principles (The 5 C's)

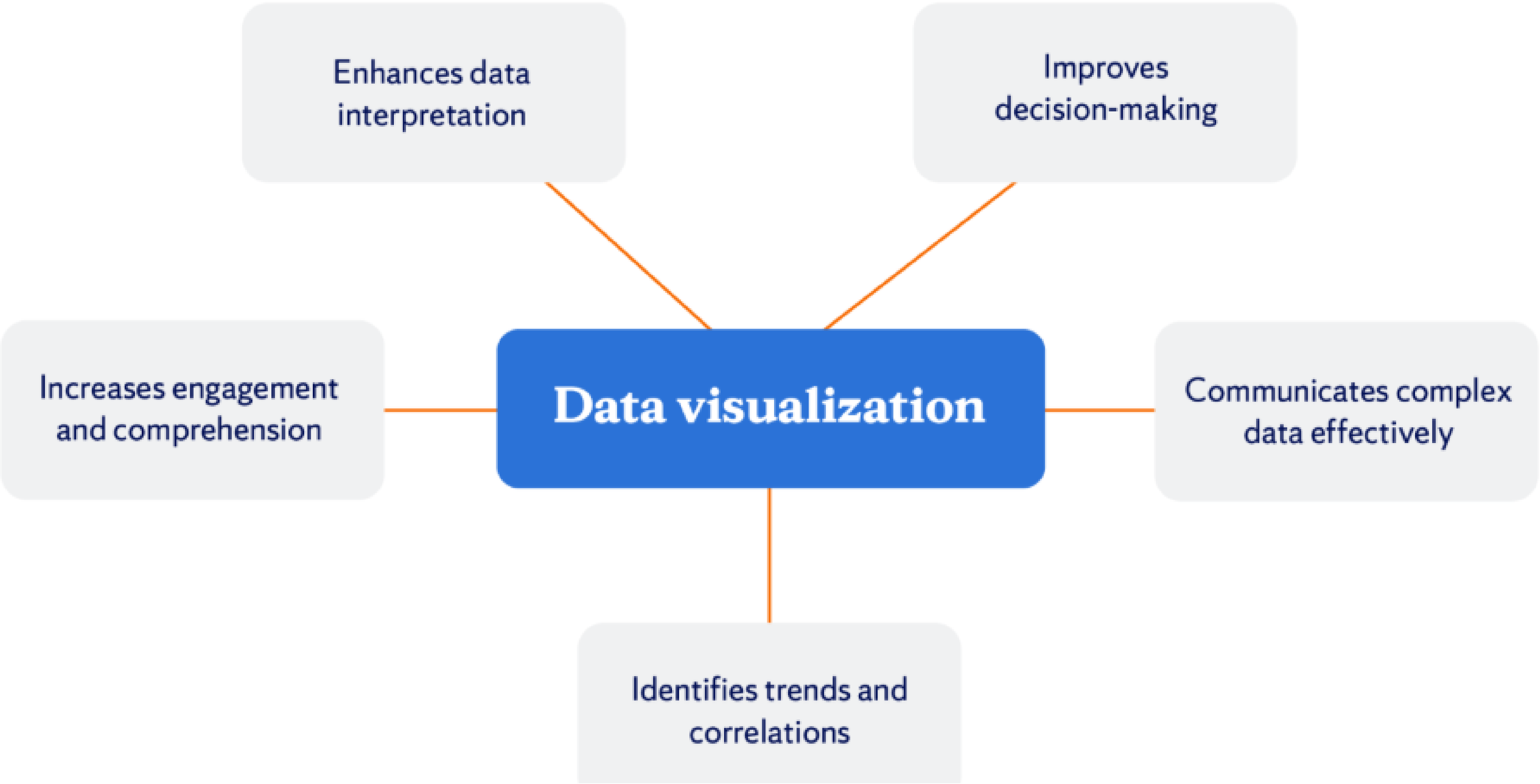
To ensure these goals are met, practitioners **follow the 5 C's of data visualization:**

- Clarity: The message should be easy to interpret without confusion.
- Context: Provide titles, labels, and metadata so viewers understand the significance of the data.
- Color: Use color intentionally to highlight important data or categorize information.
- Consistency: Maintain uniform formats and styles across a suite of visuals.
- Chart Selection: Choose the specific chart type (e.g., bar, line) that best fits the nature of the data and the analytical task.

Why Data Visualization is Important

Data visualization serves several crucial purposes:

1. **Simplifies Complex Data**: Visuals make it easier to understand patterns, trends, and outliers in data.
2. **Enhances Communication**: Visual representations are often more impactful than raw data tables or reports.
3. **Supports Decision-Making**: By presenting data in a clear and concise manner, stakeholders can make informed decisions quickly.


Data visualization
Enhances data interpretation
Improves decision-making
Increases engagement and comprehension
Communicates complex data effectively
Identifies trends and correlations

Types of data visualization

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#1: The gap between the "happys" and the "happy-nots" is enormous

The top 10 happiest countries (Finland, Denmark, Switzerland, Iceland, Norway, the Netherlands, Sweden, New Zealand, Austria, and Luxembourg) scored an average of 7.5 on the life satisfaction question above. The bottom 10 countries (Afghanistan, South Sudan, Zimbabwe, Rwanda, Central African Republic, Tanzania, Botswana, Yemen, Malawi, and India) averaged 3.3.

#2: People who move to happier countries become happier

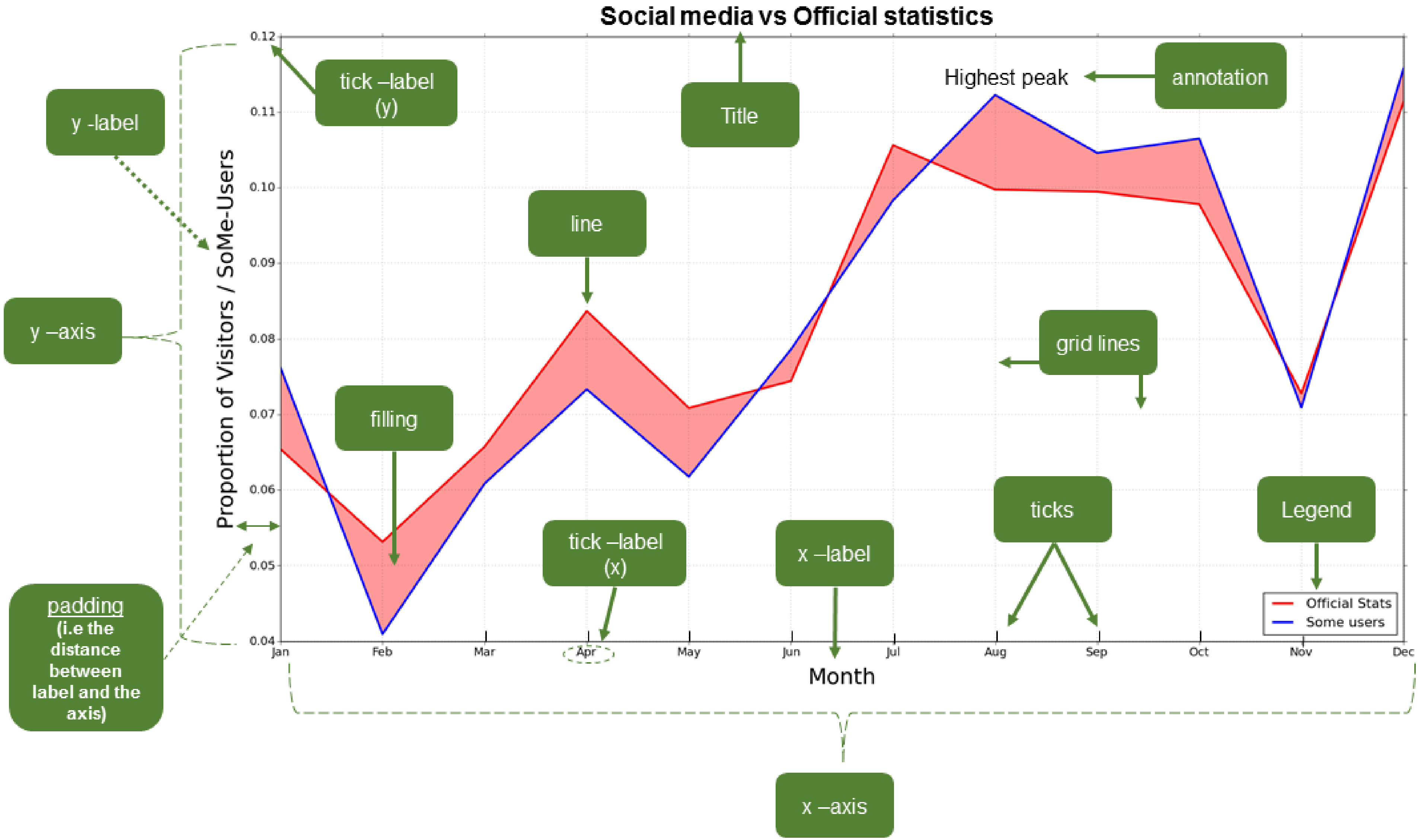
Interestingly, they found that people who move to a new country shed their old country's level of happiness and adopt their new country's level of happiness.

#3: Smaller countries are getting happier; larger countries are not

That happiness levels are improving in smaller countries and worsening in larger ones.

Even when you remove the biggest five countries (China, India, the United States, Indonesia, and Brazil) from the analysis for fear of outlier effects, the trend remains.

Anatomy of a Plot



Introduction to Matplotlib

Matplotlib is a low-level plotting library in Python that provides complete control over chart customization. Developed by John D. Hunter in 2003, it is the foundation of many other visualization libraries, including Seaborn.

Key Features of Matplotlib

- **Highly Customizable**: Allows precise control over all aspects of a plot, including colors, labels, and gridlines.
- **Supports Multiple Plot Types**: From simple line charts to advanced 3D plots, Matplotlib can handle various visualization needs.
- **Integration with Other Libraries**: Works seamlessly with NumPy, Pandas, and other Python libraries.

Matplotlib

Matplotlib is a Python library for creating static, animated and interactive visualizations.

Plot types

01 Line plot

02 Bar plot

03 Scatter plot

04 Pie plot

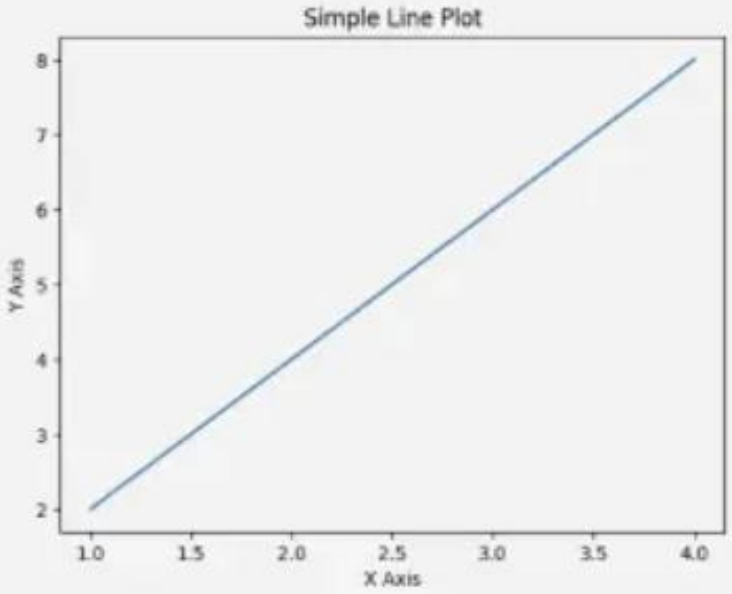
05 Histogram

Line plot

A line plot connects data points with straight lines to show trends. In Matplotlib, the plot() function is used to create it.

Common uses of line plots:

- Tracking time series data (e.g., stock prices, temperature).
- Comparing multiple datasets.
- Identifying patterns and trends in continuous data.

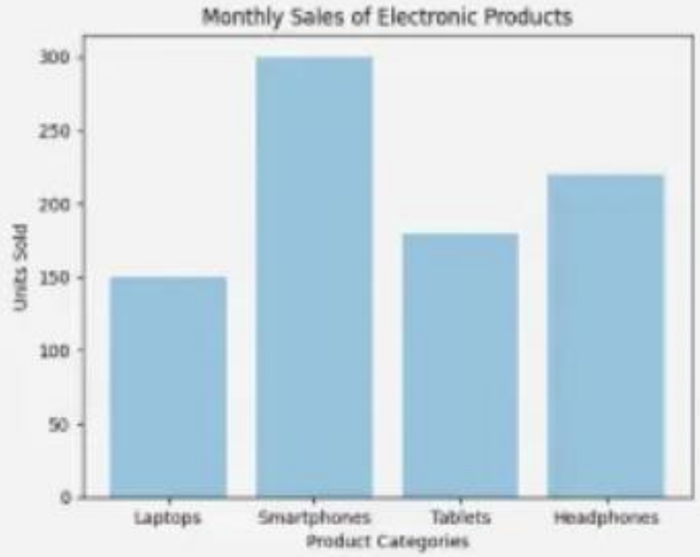


Bar Plot

A bar plot uses rectangular bars to represent categorical data, with height indicating value. In Matplotlib, it's created using the bar() function.

Common uses of bar plots:

- Comparing categories or groups.
- Showing distributions and frequencies.
- Visualizing differences in data values.

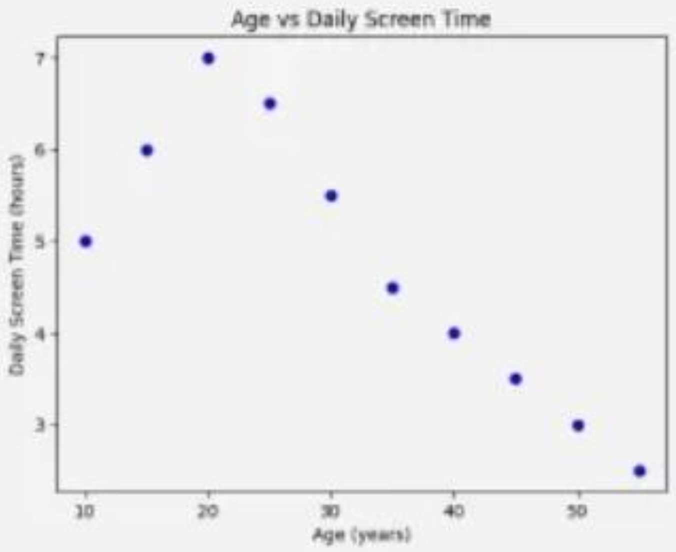


Scatter Plot

A scatter plot visualizes relationships between two variables using individual data points. In Matplotlib, it's created with the scatter() function.

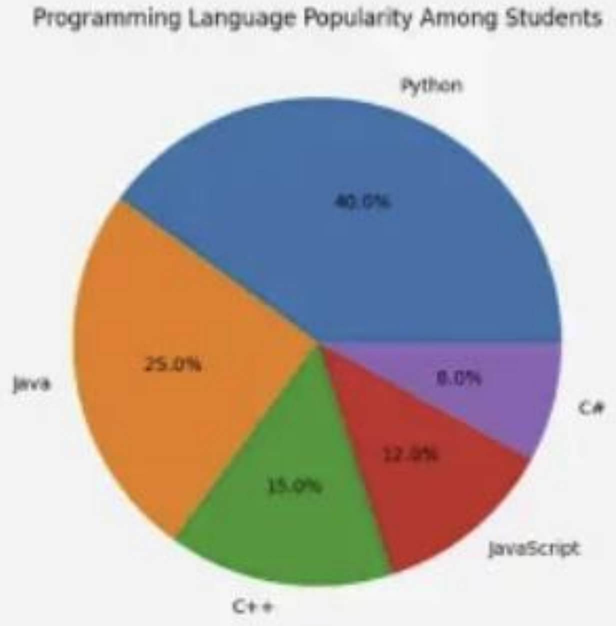
Common uses of scatter plots:

- Showing relationships between two numerical variables.
- Identifying trends, clusters, and outliers.
- Visualizing correlations in data.



Pie plot

A pie plot represents data as slices of a circle, showing proportions of a whole. In Matplotlib, it's created using the pie() function.



Common uses of pie plots:

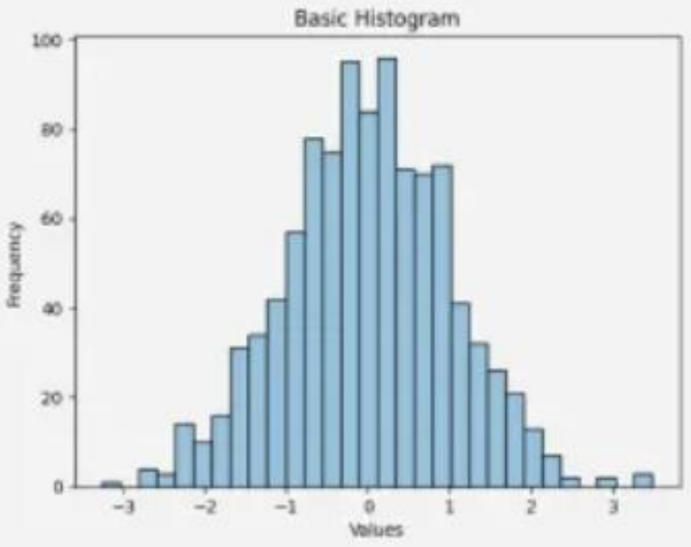
- Showing proportions or percentage distribution.
- Visualizing categorical data as parts of a whole.
- Comparing relative sizes of different categories.

Histogram

A histogram represents the distribution of numerical data using bars, where height indicates frequency. In Matplotlib, it's created using the hist() function.

Common uses of histograms:

- Visualizing the distribution of continuous data.
- Identifying patterns, skewness, and outliers.
- Understanding frequency distribution in datasets.



Example - Matplotlib

```python
import matplotlib.pyplot as plt

# Sample data
years = [2018, 2019, 2020, 2021, 2022]
sales = [250, 300, 350, 400, 450]

# Creating a line chart
plt.plot(years, sales, marker='o', color='b', label='Sales Over Years')
plt.title('Annual Sales')
plt.xlabel('Year')
plt.ylabel('Sales (in USD)')
plt.legend()
plt.grid(True)
plt.show()
```

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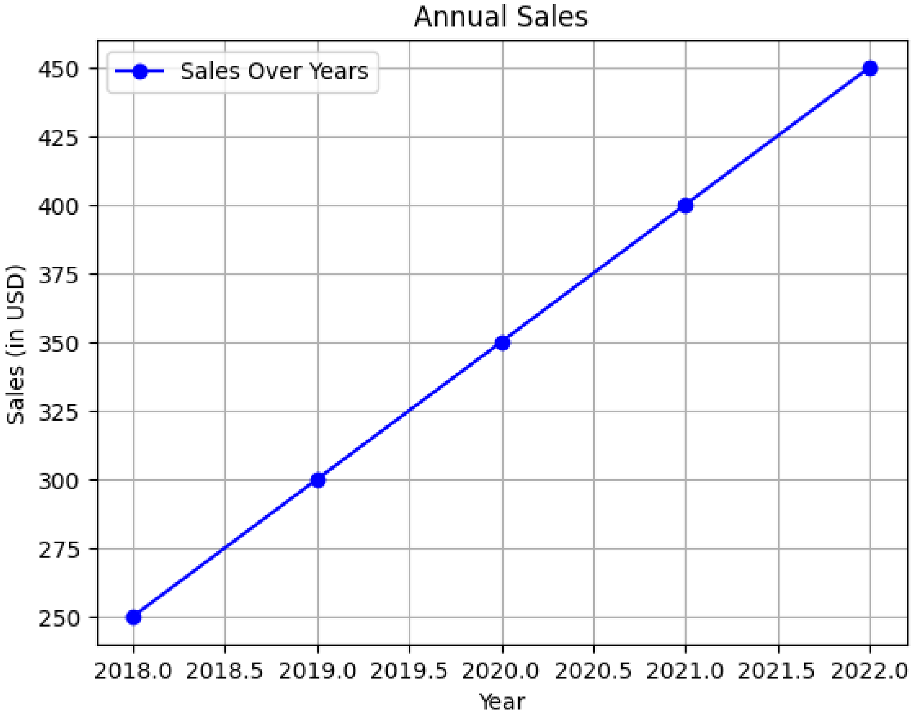


https://colab.research.google.com/github/avakanski/Fall-2024-Applied-Data-Science-with-Python/blob/main/docs/Lectures/Theme_2-Data_Engineering/Lecture_8-Matplotlib/Lecture_8-Matplotlib.ipynb

Introduction to Seaborn

Seaborn is built on top of Matplotlib and provides a high-level interface for creating attractive and informative statistical graphics. It simplifies the process of generating complex plots and is particularly useful for visualizing datasets stored in Pandas DataFrames.

Key Features of Seaborn

- **Built-in Themes**: Offers aesthetically pleasing default styles that require minimal customization.
- **DataFrame Integration**: Makes it easy to visualize data directly from Pandas DataFrames.
- **Specialized Plot Types**: Provides advanced plots such as heatmaps, violin plots, and pair plots.

Why Use Seaborn

- Provides beautiful default styles and color themes.
- Offers high-level functions for quick and clean plots.
- Integrates smoothly with Pandas for data-friendly plotting.
- Excels at relational, distribution, and category-based visualizations.

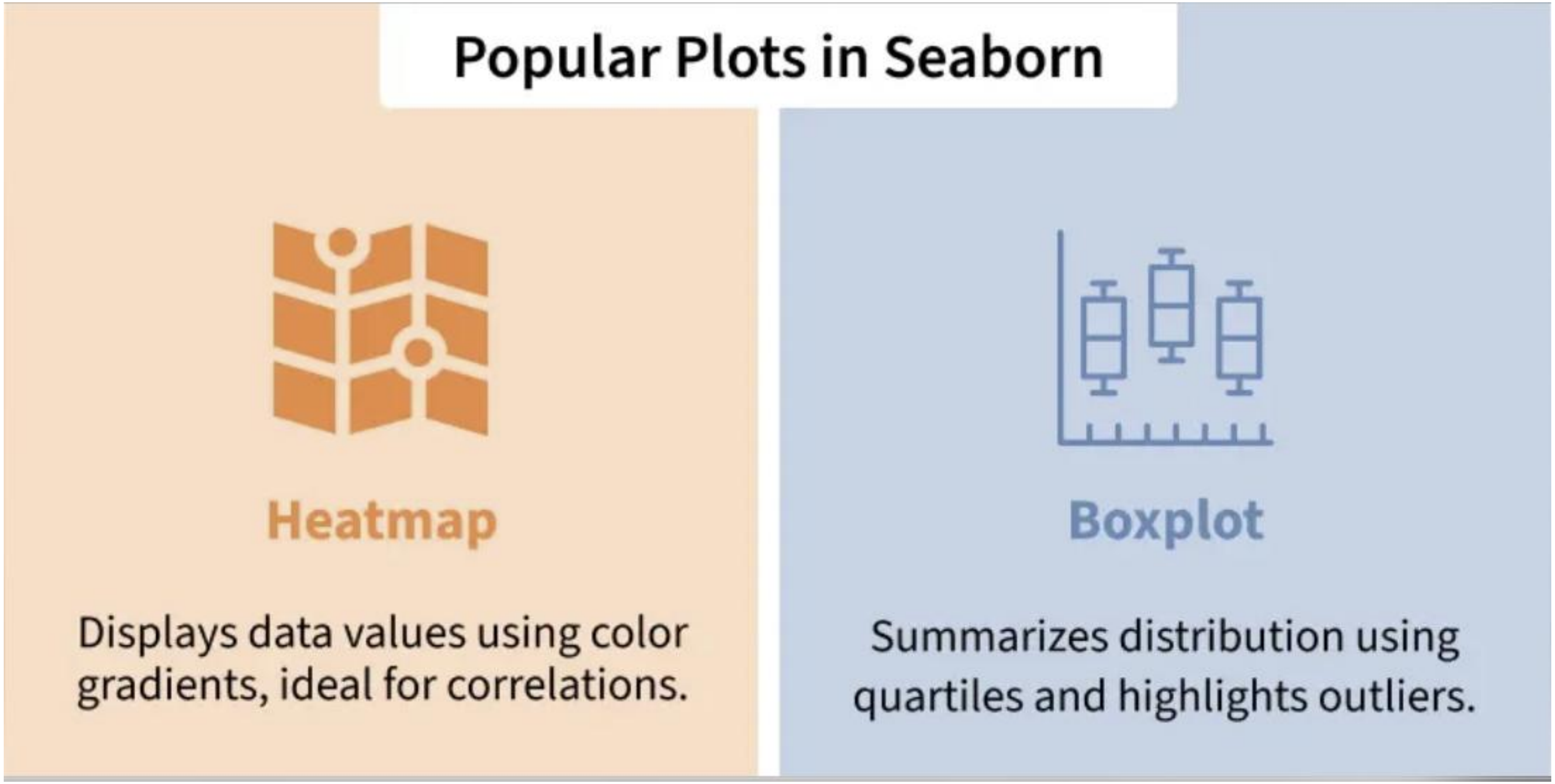
Popular Plots in Seaborn

Scatter Plot

Used to analyze the relationship between two numerical variables.

Line Plot

Great for tracking trends and changes over time or continuous data.


Popular Plots in Seaborn
Heatmap
Displays data values using color gradients, ideal for correlations.
Boxplot
Summarizes distribution using quartiles and highlights outliers.

Popular Plots in Seaborn

Bar Plot

Shows comparisons across categories using aggregated values.

Histogram / KDE Plot

Reveals the distribution, spread, and shape of your data.

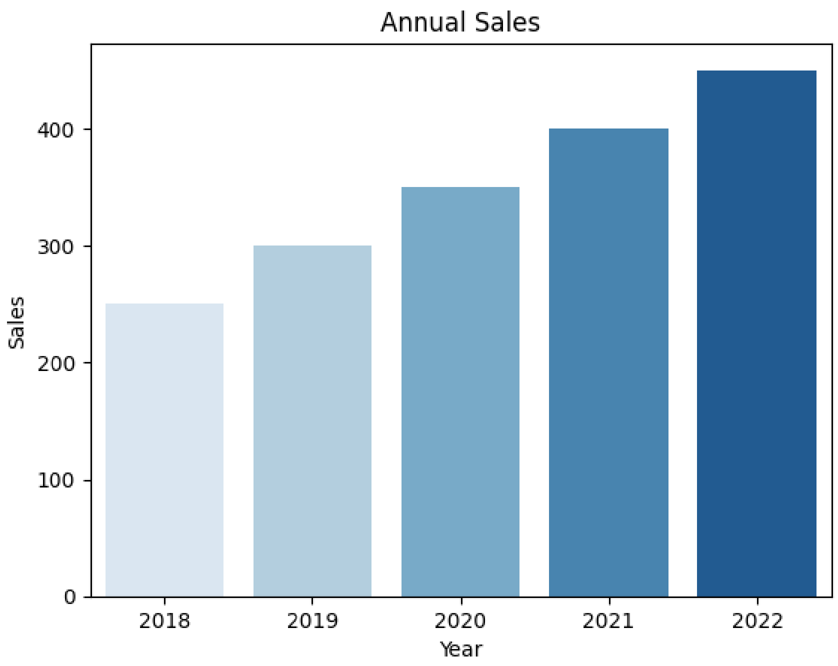
Example - Seaborn

```python
import seaborn as sns
import pandas as pd

# Sample data
data = pd.DataFrame({
    'Year': [2018, 2019, 2020, 2021, 2022],
    'Sales': [250, 300, 350, 400, 450]
})

# Creating a bar plot
sns.barplot(x='Year', y='Sales', data=data, palette='Blues')
plt.title('Annual Sales')
plt.show()
```

✓ 0.3s



https://colab.research.google.com/github/avakanski/Fall-2024-Applied-Data-Science-with-Python/blob/main/docs/Lectures/Theme_2-Data_Engineering/Lecture_9-Seaborn/Lecture_9-Seaborn.ipynb

Comparing Matplotlib and Seaborn

While both libraries are excellent for data visualization, they serve different purposes:

Feature	Matplotlib	Seaborn
Customization	Highly customizable	Limited customization
Ease of Use	Steeper learning curve	Beginner-friendly
Advanced Plots	Requires more code	Built-in advanced plot types
Style	Basic by default	Aesthetically pleasing defaults

https://colab.research.google.com/drive/1sn6sL3O8ocor8axNIWdGALnKyd2sIeCS?usp=sharing

References

Matplotlib

https://matplotlib.org/

https://www.geeksforgeeks.org/python/matplotlib-tutorial/

Seaborn

https://seaborn.pydata.org/

https://www.geeksforgeeks.org/python/python-seaborn-tutorial/

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